Exp 10
27/8/26 Internetworking with Routers in Cisco Packet Tracer

Aim :-
To implement internetworking with routers using Cisco Packet Tracer simulator.

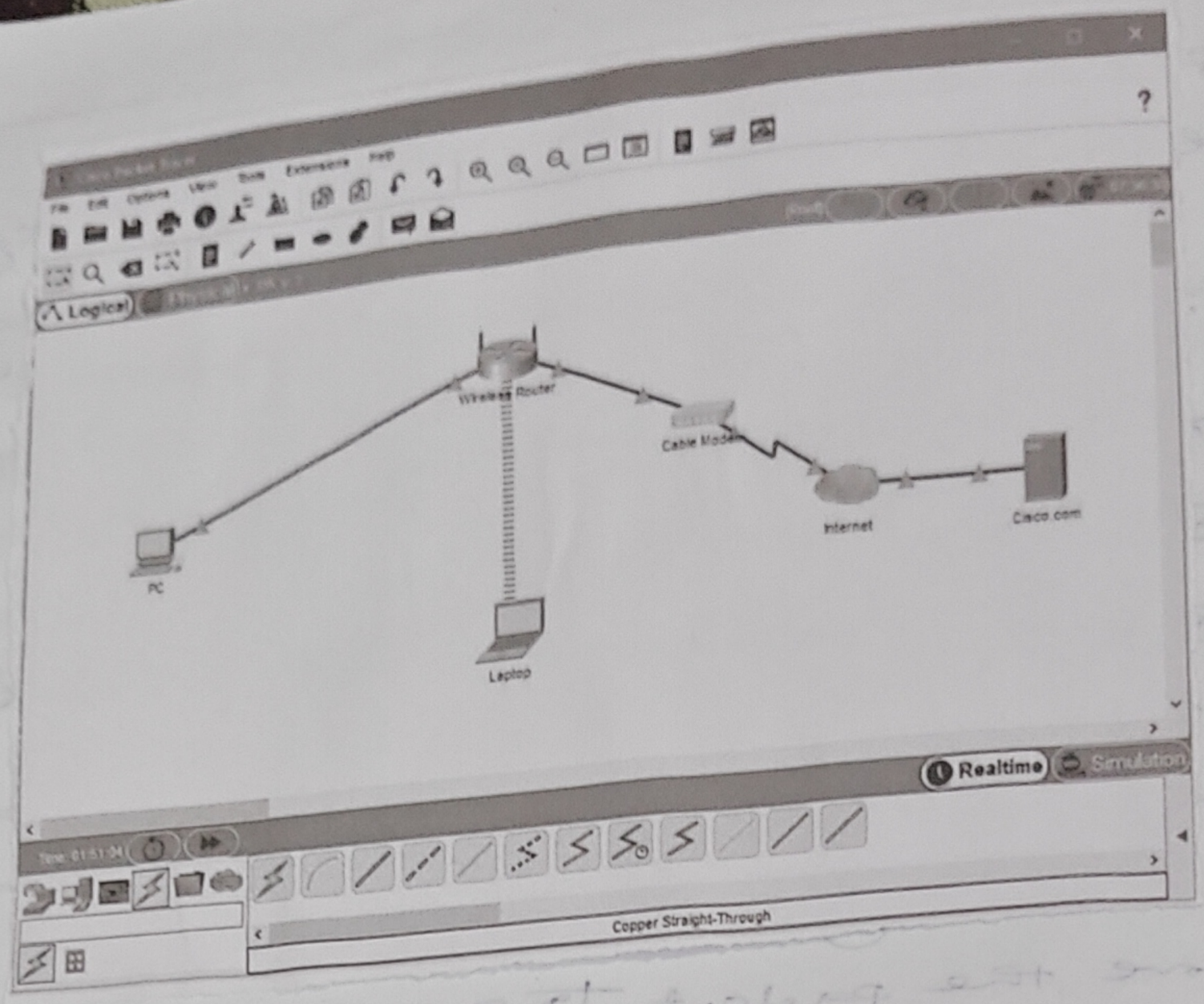
Procedure :-

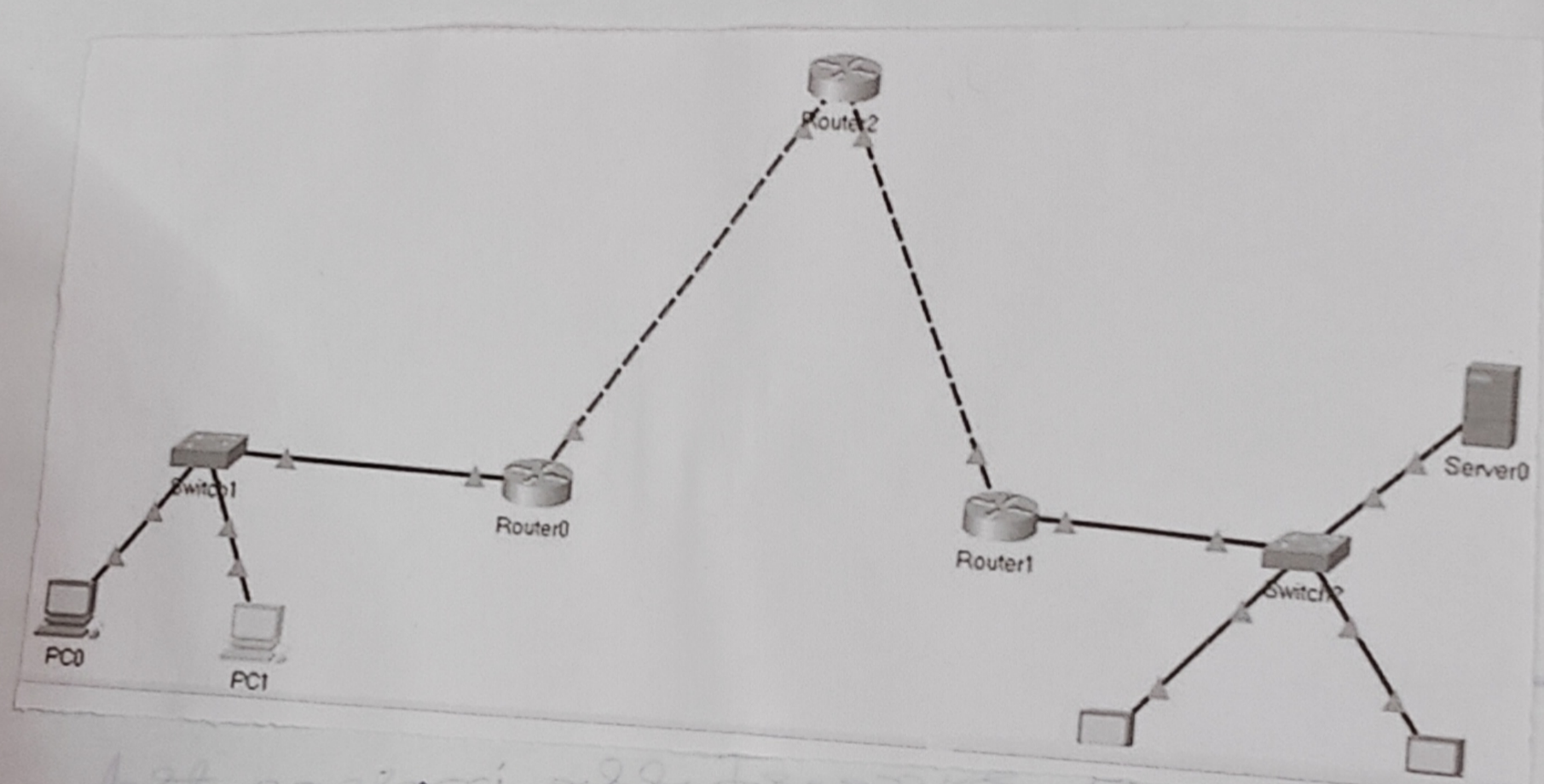
Part A - Design & configure simple internetwork using router

1. Open Cisco Packet Tracer & create network using one router & 2 PCs
2. connect PCs to router using copper straight through cables
3. open the router CLI & configure router interfaces
4. configure PCs
5. connect PC0 fastethernet to router1 fastethernet 0/0 & PC1 fastethernet to router1 fastethernet 0/1
6. perform simulation by transferring PDU from PC0 to PC1

Part B - Wireless Router, DHCP server & Internet cloud

1. Launch Packet Tracer & build topology
2. Connect PC to wireless router
3. connect wireless router to cable modem
4. connect internet cloud to cisco.com server
5. configure wireless router
6. Enable DHCP server & configure DNS server


Logical
Wireless Router
Cable Modem
Internet
Cisco.com
PC
Laptop
Realtime
Simulation
Copper Straight-Through


Router2
Switch1
Router0
Router1
Server0
PC0
PC1

If the main route is up.

```
C:\>tracert 30.0.0.20

Tracing route to 30.0.0.20 over a maximum of

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    0 ms      0 ms      0 ms      20.0.0.2
  3    *         0 ms      0 ms      30.0.0.20

Trace complete.
```

```
Router>enable
Router#show ip route static
      30.0.0.0/8 is variably subnetted, 2 s
S        30.0.0.0/8 [10/0] via 20.0.0.2
S        30.0.0.100/32 [10/0] via 40.0.0.2
S     50.0.0.0/8 [10/0] via 40.0.0.2

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Inte
GigabitEthernet0/1, changed state to down
```

If the main route is down.

```
C:\>tracert 30.0.0.20

Tracing route to 30.0.0.20 over a maximum of

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    0 ms      0 ms      0 ms      40.0.0.2
  3    0 ms      0 ms      0 ms      50.0.0.2
  4    0 ms      0 ms      0 ms      30.0.0.20

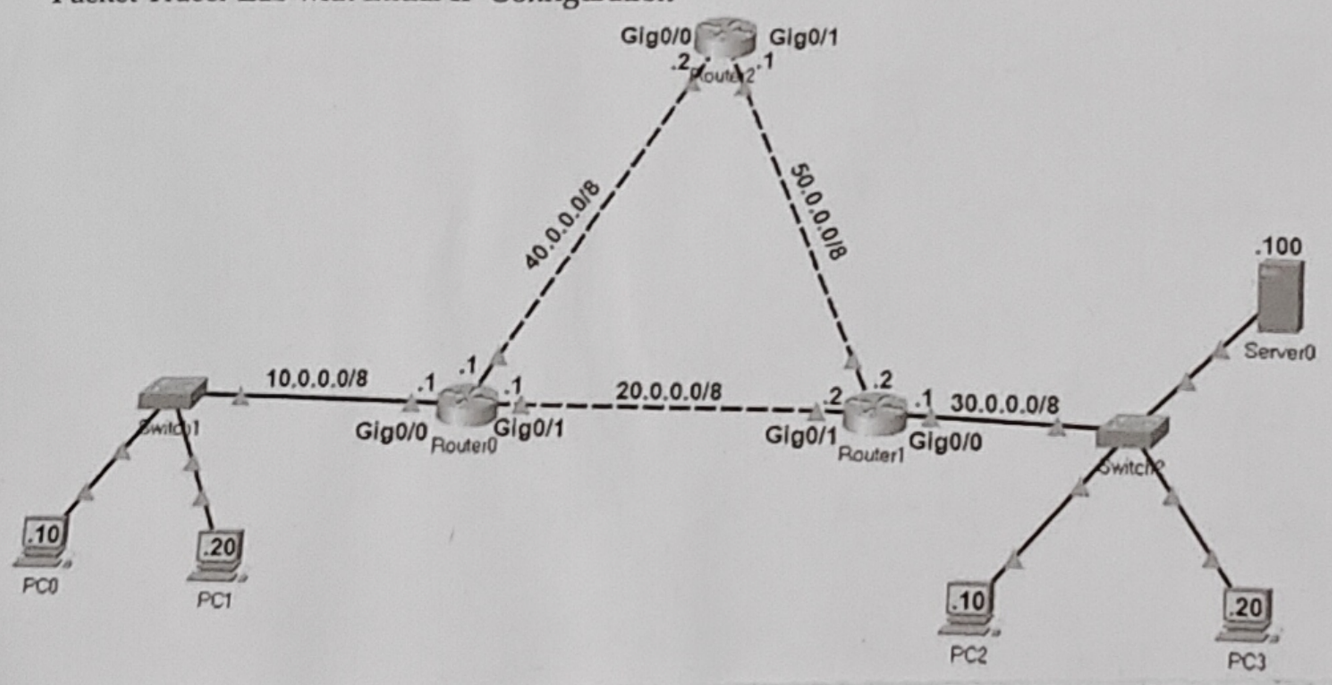
Trace complete.

C:\>
```

```
Router#show ip route static
      30.0.0.0/8 is variably subnetted, 2 s
S        30.0.0.0/8 [20/0] via 40.0.0.2
S        30.0.0.100/32 [10/0] via 40.0.0.2
S     50.0.0.0/8 [10/0] via 40.0.0.2

Router#
```

Packet Tracer Lab with Initial IP Configuration



Part C :-

1. Open PC command prompt
2. Use ipconfig / release ipconfig / renew
3. verify that PC received IP address
4. Test connectivity to cisco.com
5. observe replies received from server.

Result:

Internetwork successfully designed & configured using Cisco Packet Tracer & connectivity b/w network devices was verified.